

Table 4. Model Comparison: Nested Regression Models for Body Mass Prediction

Statistic

Linear Regression Models

Model 1 (Flipper Only)

Model 2 (+ Species)

Model 3 (+ Bill Measures)

Model 4 (Full Model)

R-squared

0.762

0.787

0.849

0.875

Adjusted R-squared

0.761

0.785

0.847

0.873

RMSE (g)

392.2

371.0

311.9

284.3

AIC

4928.1

4895.3

4783.7

4723.9

BIC

4939.6

4914.3

4810.3

4754.4

F-statistic

1060.30

405.28

369.14

380.20

df (model, residual)

(1, 331)

(2, 329)

(4, 327)

(5, 326)

p-value (F-test)

<0.001

<0.001

<0.001

<0.001

Source: Palmer Penguins Dataset (Gorman et al., 2014)

Model 1: Body Mass ~ Flipper Length

Model 2: Body Mass ~ Flipper Length + Species

Model 3: Body Mass ~ Flipper Length + Species + Bill Length + Bill Depth

Model 4: Body Mass ~ All morphometric predictors + Species + Sex